

$$W_{pay}, R, n_c; W_{total}^{(0)}, V_{\infty}^{(0)}, r^{(0)}, J^{(0)}, S_w^{(0)}$$

SLSQP Optimizer

$$W_{total}, r$$

$$W_{total}, V_{\infty}, r, S_w, J$$

$$V_{\infty}, r$$

$$W_{total}, r, S_w$$

$$R, n_c, V_{\infty}$$

$$W_{total}, W_{pay}$$

$$V_{\infty}, S_w$$

$$\min W_{total}; W_{total}^*, V_{\infty}^*, r^*, J^*, S_w^*$$

Hover Power

$$P_{hover}$$

$$P_{hover}$$

Aerodynamic Trim

$$T_{cruise}, V_i, P_0$$

Cruise Power

$$P_{cruise}$$

$$P_{cruise}$$

Installed Power

$$P_{inst}$$

Empty Weight

$$W_{empty}$$

Mission Energy

$$E_{req}$$

Battery Weight

$$W_{battery}$$

Weight Balance

$$W_{total} - W_{pay} - W_{battery} - W_{empty}$$

loading constraints

Constraints